Pooling Bias Meets Attention: Heterogeneous Panel Forecasting with Lightweight Adapters

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Abstract

Time series foundation models (TSFMs) process panel data by sharing parameters across all units, implicitly assuming homogeneous covariate responses. When units differ in their sensitivity to shared covariates—a setting well-studied in econometrics as coefficient heterogeneity—this pooling produces systematic, unit-dependent forecast bias. We formalize this failure mode by connecting the classical Pesaran & Smith (1995) pooling bias result to the TSFM setting, and propose a lightweight output-correction adapter (3,340 parameters) that attaches to a frozen TSFM backbone. The adapter takes unit-level information—either static metadata or context-inferred covariation summaries—and produces an additive forecast correction. On synthetic heterogeneous panel data, the adapter reduces test RMSE by 15–26% over the frozen base TSFM, with the largest gains on high-sensitivity units and short-context regimes where implicit adaptation fails. Ablations over heterogeneity levels and context lengths confirm that the adapter's value increases precisely when the pooling bias is most severe.

1 Introduction

Time series foundation models (TSFMs) have achieved strong zero-shot and few-shot forecasting performance across diverse domains [3, 4, 5]. A common deployment scenario involves *panel data*: many related units (customers, sensors, regions) observed over time with shared exogenous covariates. TSFMs process each unit's history through shared parameters—patch embeddings, attention layers, and prediction heads—producing forecasts that implicitly assume all units respond identically to the same covariate.

This assumption is often violated. In credit risk, different borrowers react differently to interest rate changes. In energy forecasting, buildings vary in temperature sensitivity. In retail, stores differ in price elasticity. The econometrics literature has long recognized this as *coefficient heterogeneity*: when unit-specific coefficients β_i are heterogeneous but estimated via pooled regression, the resulting estimator is inconsistent even as $N, T \to \infty$ [1]. The pooling bias is proportional to the covariate magnitude $|x_t|$, causing forecast errors to amplify precisely when covariates take extreme values.

Despite extensive work on covariate integration in TSFMs [6, 7, 8], no prior work has addressed unit-level heterogeneity in covariate response. Existing adapters treat covariate injection as a homogeneous operation—the same transformation is applied regardless of which unit is being forecast.

We make three contributions: (1) We connect the Pesaran & Smith pooling bias to the TSFM setting and demonstrate empirically that frozen TSFMs exhibit the predicted bias amplification pattern on heterogeneous panel data. (2) We propose a lightweight output-correction adapter (3,340 parameters, $<$0.4% of the backbone) that corrects the unit-specific bias using either static metadata (Variant A) or

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context-inferred covariation (Variant B). (3) We show through ablations that the adapter's value scales with heterogeneity severity and inversely with context length, consistent with theoretical predictions.

2 Method

2.1 Problem Setup

Consider a panel of N units observed over T time steps. Unit i at time t follows:

$$y_{it} = f_i(y_{i,<t}) + \boldsymbol{\beta}_i^\top \mathbf{x}_t + \varepsilon_{it}, \tag{1}$$

where f_i captures unit-specific dynamics, $\mathbf{x}_t \in \mathbb{R}^C$ is a vector of shared covariates, $\boldsymbol{\beta}_i \in \mathbb{R}^C$ is the unit-specific covariate sensitivity, and ε_{it} is noise.

A TSFM with shared parameters produces forecast $\hat{y}_{it}$ that implicitly uses a pooled coefficient $\bar{\boldsymbol{\beta}}$. By the Pesaran & Smith (1995) result, the forecast error for unit i contains a bias term proportional to $(\boldsymbol{\beta}_i - \bar{\boldsymbol{\beta}})^\top \mathbf{x}_t$, which grows with $|\mathbf{x}_t|$.

2.2 Output-Correction Adapter

Given a frozen TSFM backbone f_θ that produces base forecast $\hat{\mathbf{y}}_i^{\text{base}} \in \mathbb{R}^H$ (for horizon H), our adapter computes an additive correction:

$$\hat{\mathbf{y}}_i = \hat{\mathbf{y}}_i^{\text{base}} + g_\phi(\mathbf{s}_i, \mathbf{x}_{T+1:T+H}), \tag{2}$$

where g_ϕ is a two-layer MLP with 3,340 parameters, and $\mathbf{s}_i$ is a unit-level feature vector. The correction is initialized near zero (random Xavier initialization with small weights), so the adapter begins as a no-op and learns to deviate only where beneficial.

Variant A (Static): $\mathbf{s}_i = \mathbf{z}_i$, where $\mathbf{z}_i$ is a static metadata vector (e.g., unit type, segment indicators).

Variant B (Inferred): $\mathbf{s}_i = \frac{1}{T_c} \sum_t y_{it} \cdot \mathbf{x}_t$, the empirical covariation between the unit's history and covariates over the context window. This provides a noisy estimate of $\boldsymbol{\beta}_i$ without requiring metadata.

2.3 Training Procedure

We use a four-way temporal split to ensure the adapter learns from the backbone's out-of-sample bias rather than its in-sample residuals:

1. **Base train** [0%, 50%): Train backbone end-to-end (all parameters).
2. **Gate train** [50%, 70%): Freeze backbone, train only adapter on data where the backbone exhibits generalization bias.
3. **Validation** [70%, 85%): Select best adapter checkpoint.
4. **Test** [85%, 100%): Final evaluation.

This split is critical: on the base training data, the backbone's bias is near zero (it has already fit this data), so the adapter would learn a trivial correction. By training the adapter on held-out data where the backbone generalizes imperfectly, it learns to correct genuine systematic bias.

3 Experiments

3.1 Setup

Synthetic DGP. We generate panel data with N=2000 units, T=512 time steps, C=3 covariates, and 3 groups with heterogeneous sensitivity vectors $\boldsymbol{\beta} \in \{[0.1, 0.1, 0.2], [1.5, 1.0, 1.5], [5.0, 3.0, 5.0]\}$. The DGP includes nonlinear autoregressive dynamics ($\phi y_{t-1} + \gamma \sin(y_{t-1})$), unit-specific trends, shared seasonal components, and AR(1) covariates. We calibrate following standard Monte Carlo designs in the heterogeneous panel literature [1, 2].

Base TSFM. A patch-based Transformer (4 layers, d=128, 4 heads, 869K parameters) trained from scratch, following the lightweight TSFM paradigm of TTM [11]. Context length is 32, forecast horizon is 12.

Table 1: Test set performance. RMSE and per-group RMSE (G0: low β, G1: medium, G2: high). Improvement (%) is relative to Base TSFM.

Model	RMSE	RMSE (G0)	RMSE (G1)	RMSE (G2)	Impr.
Base TSFM	12.72	1.49	6.47	20.99	—
+ Static (A)	9.94	1.17	5.13	16.38	−21.8%
+ Inferred (B)	10.84	1.31	5.50	17.91	−14.7%
+ Oracle	9.42	1.32	5.04	15.47	−25.9%

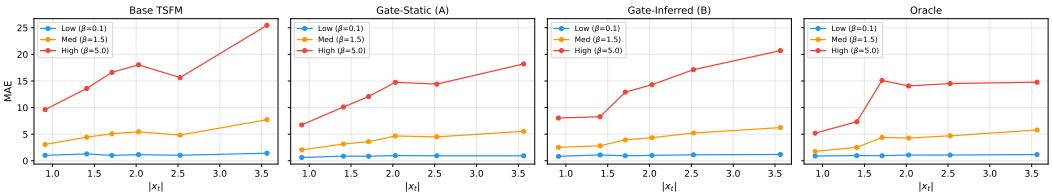

Figure 1: MAE vs. covariate magnitude $|x_t|$ by sensitivity group. Left: Base TSFM shows bias amplification—high-β group error grows with $|x_t|$. Right panels: adapters compress the inter-group gap.

Adapter. A two-layer MLP (3,340 parameters, <0.4% of backbone) attached after the frozen backbone. Trained with constant learning rate 10^{-3} and early stopping (patience 20).

3.2 Main Results

Table 1 shows the adapter substantially reduces forecast error, with the largest gains on the high-sensitivity group (G2: 21.0 → 15.5–17.9). The ordering Oracle > Static > Inferred > Base is consistent with theoretical expectations: richer unit-level information enables better bias correction. Notably, Variant B (Inferred) achieves 14.7% improvement without any metadata, using only the historical covariation pattern to estimate unit sensitivity.

Figure 1 shows the bias amplification pattern. The base TSFM exhibits the Pesaran & Smith prediction: MAE for high-β units grows with covariate magnitude $|x_t|$, while low-β units remain stable. The adapter compresses this fan-out, producing more uniform error across groups and covariate magnitudes.

3.3 Ablation: Context Length

Table 2 reveals a clear trend: adapter value decreases monotonically with context length. At context 16–32, the adapter provides 15–19% improvement. At context 64, gains shrink to 5–9%. At context 96, the base TSFM has sufficient history to implicitly infer unit-specific sensitivity through attention, and the adapter provides no additional benefit. This confirms that the adapter is most valuable in short-context regimes—precisely the cold-start and limited-history scenarios common in real-world panel deployment.

3.4 Ablation: Heterogeneity Level

Table 3 shows the adapter improves performance across all heterogeneity levels (17–26%), confirming its robustness.

4 Related Work

TSFM covariate handling. CoRA [6] injects covariates via learned Granger Causality Embedding into frozen TSFMs. ChronosX [7] fine-tunes MLP adapter layers for covariate integration. TF-MAdapter [8] cascades univariate TSFM forecasts with covariates via non-parametric instance-level learning. None of these address unit-level heterogeneity in covariate response.

Table 2: Adapter improvement (% RMSE reduction vs. base) across context lengths. Shorter context $\rightarrow$ weaker implicit adaptation $\rightarrow$ larger adapter value.

Context	Base RMSE	Static (%)	Inferred (%)
16	10.87	-15.8	-18.9
32	12.72	-14.8	-18.0
64	11.04	-5.0	-8.6
96	9.88	$+8.4$	$+0.6$

Table 3: Adapter improvement across heterogeneity levels (primary covariate β shown). Gate improvement is substantial across all levels.

Level	β spread	Base RMSE	Static (%)	Inferred (%)
Low	[0.8, 1.0, 1.2]	4.65	-19.3	-26.1
Med	[0.5, 1.5, 3.0]	8.30	-25.5	-24.2
High	[0.1, 1.5, 5.0]	12.72	-18.8	-17.6

TSFM adaptation. δ-Adapter [9] proposes input-nudging and output-correction placements for frozen forecasters, closest to our approach. TS-Memory [10] distills retrieval-based correction into a lightweight adapter. Our work differs by providing an econometric motivation for *why* correction is needed and *what form* it should take.

Heterogeneous panels in econometrics. Pesaran & Smith [1] proved that pooled estimators are inconsistent under coefficient heterogeneity. Bonhomme & Manresa [2] proposed grouped fixed-effects estimators. We bridge this literature to the TSFM setting.

5 Discussion

We demonstrated that coefficient heterogeneity—a well-understood econometric phenomenon—creates systematic forecast bias in TSFMs, and that a lightweight output-correction adapter can substantially mitigate it. The adapter is model-agnostic, requiring no modification to the TSFM backbone, and adds less than 0.4% parameters.

A key finding is the interaction between context length and adapter value: with sufficient context ($\geq$96 steps), transformer attention can implicitly infer unit-level sensitivity, but in short-context regimes, explicit correction is essential. This has practical implications for cold-start and limited-history scenarios common in finance, healthcare, and IoT applications.

Limitations. Our experiments use synthetic data; real-world panel datasets with known heterogeneous covariate effects would strengthen validation. The adapter's improvement depends on the severity of heterogeneity and the base model's inability to implicitly handle it.

Future work. Extending to real credit risk and energy panels, integrating with production TSFMs (Chronos-2, Moirai), and developing theoretical guarantees for the correction's consistency.

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